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Design, Modeling, and Validation of a High-Speed Rotary Pulse-Width-Modulation On/Off Hydraulic Valve

Efficient high-speed on/off valves are an enabling technology for applying digital control techniques such as pulse-width-modulation (PWM) to hydraulic systems. Virtually variable displacement pumps (VVDPs) are one application where variable displacement functionality is attained using a fixed-displacement pump paired with an on/off valve and an accumulator. High-speed valves increase system bandwidth and reduce output pressure ripple by enabling higher switching frequencies. In addition to fast switching, on/off valves should also have small pressure drop and low actuation power to be effective in these applications. In this paper, a new unidirectional rotary valve designed for PWM is proposed. The valve is unique in utilizing the hydraulic fluid flowing through it as a power source for rotation. An unoptimized prototype capable of high flow rate (40 lpm), high speed (2.8 ms transition time at 100 Hz PWM frequency), and low pressure drop (0.62 MPa), while consuming little actuation power (<0.5% full power or 30 W, scavenged from fluid stream), has been constructed and experimentally validated. This paper describes the valve design, analyzes its performance and losses, and develops mathematical models that can be used for design and simulation. The models are validated using experimental data from a proof-of-concept prototype. The valve efficiency is quantified and suggestions for improving the efficiency in future valves are provided. [DOI: 10.1115/1.4006621]

1 Introduction

Traditional means of controlling fluid power systems such as proportional valves and variable displacement pumps (VDPs) have their limitations. Valve controlled systems (for example, the bleed off circuit in Fig. 1) are typically compact, inexpensive, and provide good control bandwidth. However, these traits come at the cost of efficiency since all excess flow is throttled by the proportional valve. In contrast, VDPs offer better efficiency since the displacement of the pump can be tuned to the load, thus producing only the required flow. The disadvantage of VDPs is that conventional electronic displacement control (EDC) piston pumps typically require 3–4 times the volume and weight of a fixed-displacement gear pump of equal displacement [1]. The added complexity of the EDC unit also increases the cost of VDPs, and the bandwidth is typically lower due to the moving mass of the displacement varying mechanism (for example, a swash plate).

An efficient alternative that retains the simplicity of valve control is the VVDP concept (see Fig. 2) [1,2]. This approach, which is the hydromechanical analogue of a switched-mode dc–dc converter, attains flow control with a fixed-displacement pump by quickly switching the output flow between a high pressure (load) branch and a low pressure (tank) branch instead of restricting the flow using an orifice. If the switching between the load and the tank is pulse-width-modulated, the mean output flow of the pump/valve system is controlled by varying the PWM duty ratio, or fraction of each period that the on/off valve is open to load (hence the use of "virtually variable" to describe the system's displacement). This approach is efficient relative to throttling because the valve loss in either the fully on or fully off state is small.

The VVDP is an example of the emerging field of digital hydraulics that necessitates advancements in on/off valve technology. Other examples include (but are not limited to) piston-by-piston digital-displacement pumps, where one or more active valves are used to control each piston [3]. Binary sequencing has also been explored, whether applied to the orifice area of a parallel array of on/off valves for approximating the function of a proportional valve (with reduced cost) [4] or to a stepped area piston (with each step activated by an on/off valve) to enable variable area pistons that can be used as transformers [5].

Operating VVDPs at high frequencies is desirable because output ripple is reduced and system bandwidth is improved [1,2]. This requires high-speed valves which have low pressure drop, high flow capacity, and low actuation power. Conventional on/off valves rely on the alternating linear motion of a spool or poppet. Therefore, increasing their flow capacity and speed requires higher actuation power. The actuation power required to overcome the inertial forces alone is proportional to the cube of the PWM frequency and to the square of the valve travel (i.e., orifice opening). High switching speeds (0.1–1.5 ms transitions) have been achieved using piezo-electric actuators [6,7], although these valves typically have limited flow capacity (8 lpm) or exhibit high pressure drop (up to 10 MPa). Solenoid based valves have been proposed which exhibit similar characteristics [8].

To overcome the fundamental trade off between valve speed, flow area, and actuation power in PWM applications, a rotary three-way on/off valve is proposed such that the on/off sequence is embedded in the continuous unidirectional rotary motion of the spool. The PWM frequency is proportional to the spool's rotational speed. This eliminates the need for the valve to start and stop so that actuation power needs only to overcome friction, which is proportional to the square of the frequency. In addition, the valve transition time as a proportion of the PWM period is fixed so that transition losses do not increase as PWM frequency increases. Since the spool has a helical shaped land on its surface, the duty ratio of the valve is determined by the axial motion of the spool in relation to fixed ports on the valve sleeve. The valve is also self-spinning: it scavenges power from the fluid flow to achieve its rotary motion so that an external rotary actuator is not needed.

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Contributed by the Dynamic Systems Division of ASME for publication in the Journal of Dynamic Systems, Measurement, and Control. Manuscript received December 31, 2009; final manuscript received April 11, 2012; published online September 13, 2012. Editor: J. Karl Hedrick.

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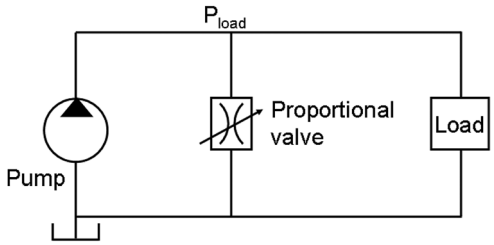


Fig. 1 Bleed off circuit

A review of the prior art has revealed several other high-speed valve designs based on rotary motion. Cui et al. [9] developed a bidirectional two-stage rotary spool valve. This valve is not continuously rotating as in the present paper. Instead, a step rotary motion of the pilot is used to create an axial pressure imbalance that causes linear sliding motion of a second stage poppet. This valve achieved a fast transition time of 2.5 ms with a pressure drop of 9 MPa at 18 lpm of flow due to a relatively small valve opening.

Continuous rotational motion, which is exploited by the valve proposed in the present paper, has been used by other research groups as well [10–14]. Royston and Singh [12] developed a pneumatic rotary PWM valve capable of 80 Hz based on a continuously rotating inner shaft with supply and return ports on a fixed outer stator. The duty ratio is set by the angular position of the load port relative to the stator. Brown et al. [10] proposed a design capable of 500 Hz PWM frequency, consisting of a concentric control shaft, hollow rotor, and stator. The rotor sets the PWM frequency while the angle between the control shaft and the stator determines the duty ratio. Recently, Van de Ven and Katz [13] proposed a design based on two tiers of continuously rotating disks designed for a PWM frequency of 100 Hz with the duty ratio determined by the phase difference between the disks. The two-way valve proposed by Cyphelly and Langen [14] uses a continuously rotating spool and a helical land that alternately opens and closes the valve, similar to what is proposed in the present paper, but it is motor driven rather than self-spinning.

The present paper is unique, however, in that it is a three-way design that alternately sends flow to either tank or the application. More importantly, it scavenges power from the fluid flow itself for rotating the spool. An unoptimized prototype capable of high flow rate (40 lpm), high speed (100 Hz PWM frequency or 2.8 ms transition time when self-spinning), and low pressure drop (0.62 MPa) has been constructed and experimentally tested. The valve consumes relatively little actuation power (30 W or less than 0.5% full power) that is scavenged from the fluid flow. This paper describes the design and also develops design equations that characterize the primary valve losses and operating characteristics. A detailed simulation and experimental study of the efficiency characteristics of the valve when used in a VVDP is also provided.

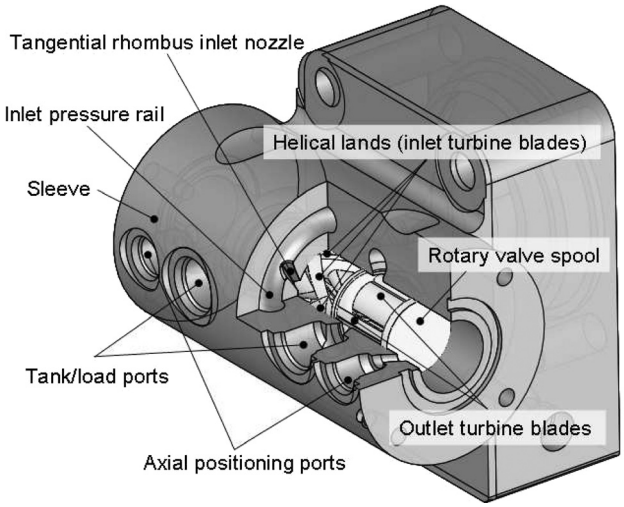


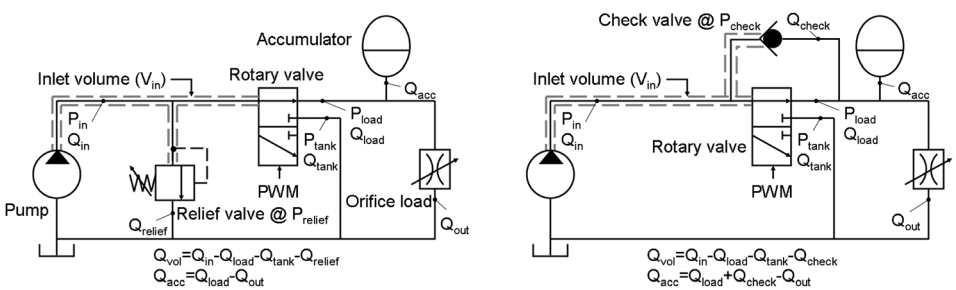
Fig. 3 Rotary valve spool/sleeve assembly. The spool rotates and translates within the sleeve bore.

Section 2 describes the three-way self-spinning rotary valve concept. The operating principle of the VVDP is outlined in Sec. 3. Design equations and loss analysis of the valve are presented in Sec. 4. A dynamic model of the VVDP is developed in Sec. 5 and an overview of the test bench is given in Sec. 6. Experimental data from an unoptimized proof-of-concept prototype are used to validate the model in Sec. 7. The validated model is then employed to predict the VVDP efficiency for an optimized valve and to compare it with a similar proportional valve controlled system. Conclusions are summarized in Sec. 8.

2 Rotary Valve Concept

The three-way rotary valve concept is illustrated in Figs. 3 and 4. The design consists of a stationary sleeve and a rotating/translating spool. The sleeve (Fig. 3) is designed to replace part of the pump housing in order to minimize the dead volume between the pump outlet and the spool. The outlet flow of the pump is directed to tangential nozzles that port fluid to the inner diameter of the sleeve bore and into the inlet section of the spool. The cross sections of the nozzles are rhombic shaped to match the helical land as this minimizes the on/off transition time for a given flow area (Fig. 5).

The spool, shown in Fig. 4, rotates and translates within the sleeve bore. The inlet (center) section of the spool is partitioned into two flow paths (highlighted in dark and light gray in Fig. 4) by overlapping helical lands. Internal pathways along the axis of the spool direct flow from the inlet section to one of the two adjacent outlet turbines. The dark gray region of the inlet section connects to the load outlet turbine (also in dark gray), and the light gray region connects to the tank outlet turbine (also in light gray).



(a) Relief configuration (b) Check configuration

Fig. 2 Two VVDP implementations. Q_{vol} and Q_{acc} represent the net flows into the inlet volume and accumulator.

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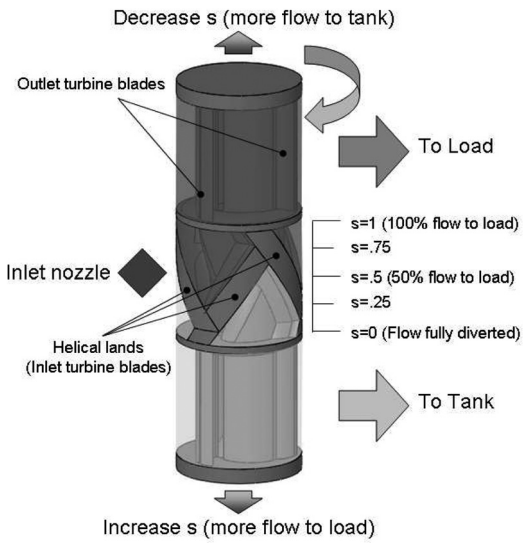

Fig. 4 Three-way helical spool concept. Internal passages connect the center section (responsible for PWM) to one of the two adjacent outlet turbines. The dark gray portions of the spool are hydraulically connected and permit flow from the inlet to the load. Similarly, the light gray sections connect the inlet to tank.

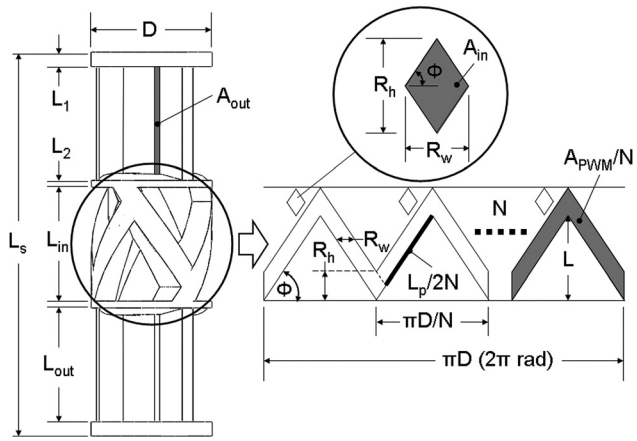

Fig. 5 2D representation of the rotary valve's geometry including variable definitions used in Sec. 4

Flow is directed from the outlet of the pump to load or tank depending on which region of the spool is connected to the ports on the sleeve. As the spool rotates, fluid is alternately ported between load and tank, thereby achieving PWM. The inlet section of the spool is divided into N sections coinciding with N inlets (see Fig. 5). Each section performs one complete on/off cycle and the PWM frequency of the valve is N times the spool's rotational frequency.

The duty ratio of the valve, s, is dependent on the spool's axial position and controls the output flow fraction of the VVDP. At $s=0$ or the zero duty ratio position (as shown in Fig. 4), flow is bypassed to tank during the entire revolution, thus sending no flow to load (i.e., no VVDP output flow). The opposite occurs at $s=1$ or full duty ratio position when the VVDP flow equals the full pump flow. Because of the spool's helical structure, the duty ratio and normalized output flow of the VVDP vary linearly with axial position. The spool's axial position is controlled using an electrohydraulic gerotor pump in a hydrostatic configuration, although other actuation methods are possible. The gerotor ports fluid to either side of the spool using the axial positioning ports shown in Fig. 3. Sensing of the axial and angular positions of the spool is accomplished using noncontact optical sensors [15,16].

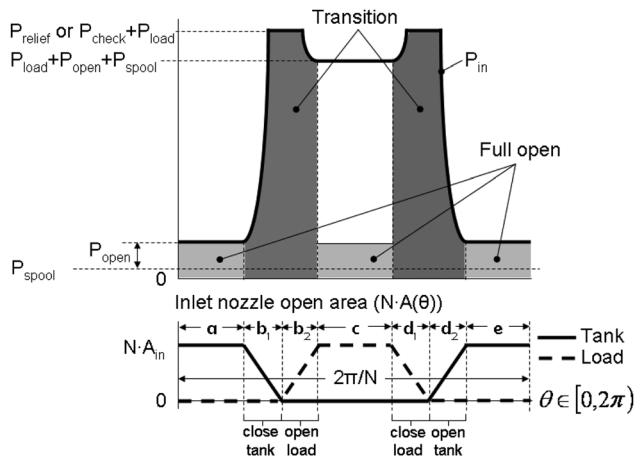

Fig. 6 Top: Inlet pressure (P_{in}) profile over one PWM cycle ($\frac{2\pi}{N}$ rad spool rotation) for the two circuits in Fig. 2. Bottom: Corresponding profiles of the valve inlet nozzle open areas ($A(\theta)$) to load or to tank.

Self-spinning of the spool is accomplished by designing both the inlet and outlet sections of the spool as turbines to capture throttling energy and fluid momentum. The inlet section acts as an impulse turbine, where fluid is accelerated via nozzles tangential to the rotor (i.e., valve spool) [17]. As the fluid impinges on the turbine blades, angular momentum is transferred to the spool as it redirects flow to the center of the spool. Fluid then exits the inlet section axially with no angular momentum. In contrast, the outlet section is designed as a reaction turbine. Blades on the outlet section guide the fluid from the center of the spool outwardly and tangentially, thus imparting a reaction torque on the spool.

3 VVDP Overview

Figure 2 illustrates two possible VVDP implementations using a three-way on/off valve. The systems differ in how the inlet pressure (P_{in}) is limited when the on/off valve is transitioning. A relief valve set to P_{relief} is used in Fig. 2(*a*), and a check valve in parallel with the on/off valve with cracking pressure P_{check} is used in Fig. 2(*b*). The check valve circuit is potentially more efficient by reducing losses during transition in two ways: (1) limiting throttling losses to P_{check} above P_{load} in comparison to a fixed relief pressure which needs to be higher than any conceivable P_{load} and (2) porting the high pressure bypassed flow to load instead of dumping the fluid to tank. The latter achieves a *soft-switching* function that reduces losses when the on/off valve is switching [18,19].

Figure 6 illustrates the inlet pressure profile of the VVDP over one PWM cycle along with the corresponding nozzle open area profiles to tank (solid) or to load (dotted). Because the rhombic orifice is designed to match the helical land geometry, the orifice opening varies linearly with the spool rotation until the orifice is completely open or closed. The full pump flow, Q_{in}, is supplied to the inlet volume V_{in} (Fig. 2) throughout the entire PWM cycle. The pressure drop across the rotary valve consists of the pressure drop across the rhombic orifices and the pressure drop across the rest of the spool. Let P_{open} denote the fully open orifice pressure drop and P_{spool} denote the pressure drop for the rest of the spool, both with the full pump flow Q_{in} through the valve. Assume that $P_{tank}=0$ for simplicity.

Interval a in Fig. 6 represents the VVDP in the off state with the rotary valve completely open to tank. Since the pump is unloaded, P_{in} is low ($=P_{open}+P_{spool}$). The accumulator supplies flow to the load and Q_{in} is returned to tank.

Interval b is when the valve transitions from being open to tank to being open to load. First, the rhombic nozzle orifice closes to tank, causing P_{in} to rise (interval b_1). Once P_{in} reaches P_{relief}

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(in Fig. 2(*a*)) or $P_{\text{check}}+P_{\text{load}}$ (in Fig. 2(*b*)), the relief or check valve opens to regulate P_{in} at these levels. Since the valve is critically lapped (i.e., the helical lands and the rhombic orifices have equal widths), the instant the orifice closes completely to tank, it begins to open to load (interval b_2). The relief or check valve will close once the orifice is sufficiently open, thus causing P_{in} to decrease.

Interval c represents the period when the valve is completely open to load. The system is in the on state and the pressure drop across the rotary valve ($P_{\text{in}}-P_{\text{load}}$) becomes $P_{\text{open}}+P_{\text{spool}}$. Q_{in} from the pump is supplied to the accumulator and the load.

Interval d is when the valve transitions from being open to load to being open to tank. As the orifice closes to load (interval d_1), P_{in} rises until the relief or check valve opens. As the orifice is fully closed to load, it begins to open to tank (interval d_2). The relief or check valve closes once the orifice is sufficiently open. As the valve becomes fully open to tank, $P_{\text{in}}=P_{\text{open}}+P_{\text{spool}}$ and the system returns to the off state.

4 Design and Performance Analysis

An analysis and design equations for the rotary valve and the two VVDP configurations shown in Fig. 2 are presented in this section. Variables used in the analysis are defined in Fig. 5. Several simplifying assumptions are applied: (1) The fixed-displacement pump is assumed to be an ideal flow source with constant output flow Q_{in}. (2) The system pressure is assumed to be capped at P_{relief} for the relief circuit and $P_{\text{check}}+P_{\text{load}}$ for the check circuit when the relief valve or check valve is activated. (3) Variation of P_{load} (load/accumulator pressure) is assumed to be slow such that it can be considered a constant. (4) Tank pressure, P_{tank}, is assumed to be zero. Dynamic effects of the inlet and accumulator pressure will be discussed later in Sec. 5.

Section 4.1 investigates throttling losses. Section 4.2 examines compressibility losses. Valve leakage is explored in Sec. 4.3. The spool's self-spinning velocity is analyzed in Sec. 4.4 and the relationship between its axial position and the VVDP's output flow is studied in Sec. 4.5. Various design trade offs and a summary of the design are discussed in Sec. 4.6.

4.1 Valve Throttling Losses. Throttling losses through the rotary valve include sleeve losses from the variable rhombic nozzle orifices, losses through the spool itself, and losses due to the relief valve or check valve. Section 4.1.1 considers the losses when the rhombic orifices are fully open and Sec. 4.1.2 considers the losses when the valve is in transition.

Let the fraction of time that the valve is in transition or fully open be κ and $(1-\kappa)$, respectively. Assuming that the duty ratio s and the spool angular frequency ω are constant while the valve is transitioning, the helical land traverses the width of the rhombus R_w (Fig. 5) during each transition. The corresponding duration of each of the four transitions is

$$T_{\text{tran}}=2R_w/(D\omega) \tag{1}$$

and the PWM period is $2\pi/(\omega N)$.

Thus

$$\kappa=\frac{4NR_w}{\pi D} \tag{2}$$

4.1.1 Full Open Orifice Losses. When the rotary valve is fully open, the pressure drop across the N rhombic orifices is described by the orifice equation [20]

$$P_{\text{open}}=\frac{\rho}{2}\left(\frac{Q_{\text{in}}}{C_dNA_{\text{in}}}\right)^2 \tag{3}$$

ρ is the density of hydraulic oil, C_d is the orifice discharge coefficient (assumed constant), and $A_{\text{in}}=0.5R_wR_h$ is the cross-sectional area of one inlet with R_w and R_h being the rhombus width and height (see Fig. 5).

The pressure drop across the spool itself, P_{spool}, is estimated using a computational fluid dynamics (CFD) generated semi-empirical formula [21] based on a scaled version of the prototype's geometry. For a given diameter (D) and flow rate (Q_{in}), P_{spool} is assumed to be constant regardless of whether the valve is connected to load or tank due to the symmetry of the flow paths. Thus, the rotary valve's full open power loss is

$$\Pi_{\text{open}}=(1-\kappa)(P_{\text{open}}+P_{\text{spool}})Q_{\text{in}} \tag{4}$$

4.1.2 Transition Losses. The transition throttling losses derived in this section include losses from blocking the inlet rhombic orifices, losses through the relief valve (Fig. 2(*a*)) or check valve (Fig. 2(*b*)), and losses through the spool. During each PWM cycle, the rotary valve undergoes four transitions: opening and closing to tank with equal energy losses (E_{tank}) and opening and closing to load also with equal energy losses (E_{load}).

For the relief circuit in Fig. 2(*a*), consider first the opening to tank transition (interval d_2 in Fig. 6). At the beginning of the transition, the rhombic inlet orifice is blocked and all flow goes through the relief valve to tank and $P_{\text{in}}=P_{\text{relief}}$. As the rhombic orifice opens to tank (with the relief valve still open), flow passes through both the relief and the on/off valve to tank with a pressure drop of P_{relief}. When the valve is sufficiently open, P_{in} falls below P_{relief} and all flow goes through the on/off valve to tank causing a pressure drop of P_{spool} across the spool.

Considering the problem in angle coordinates with $\Theta=\omega t$ where $\Theta=0$ corresponds to the start of transition, let $P_i(\Theta)$ be the instantaneous pressure drop across the orifice with varying orifice area if the full flow Q_{in} passes through it

$$P_i(\Theta)=\frac{\rho}{2}\left(\frac{Q_{\text{in}}}{C_dNA(\Theta)}\right)^2=P_{\text{open}}\left(\frac{\Theta_{\text{tran}}}{\Theta}\right)^2 \tag{5}$$

where the definition in Eq. (3) has been employed and $A(\Theta)$ is the instantaneous open area of one rhombic inlet and Θ_{tran} is the total rotational angle for each transition

$$A(\Theta)=\frac{R_hD}{4}\Theta=A_{\text{in}}\frac{\Theta}{\Theta_{\text{tran}}}\qquad(0\le\Theta\le\Theta_{\text{tran}}) \tag{6}$$

$$\Theta_{\text{tran}}=2R_w/D \tag{7}$$

Let Θ_{crit} be the critical angle when the relief valve just begins to open or close coinciding with $P_i(\Theta_{\text{crit}})=P_{\text{relief}}$. From Eqs. (5) and (6), the relationship between them is as follows

$$\frac{\Theta_{\text{crit}}}{\Theta_{\text{tran}}}=\sqrt{\frac{P_{\text{open}}}{P_{\text{relief}}}} \tag{8}$$

Therefore, considering whether the relief valve is open or closed and utilizing Eqs. (5)–(8), the resulting energy loss during each tank transition is

$$E_{\text{tank}}=\frac{Q_{\text{in}}}{\omega}\left[\underbrace{\int_0^{\Theta_{\text{crit}}}P_{\text{relief}}d\Theta}_{\text{relief open}}+\underbrace{\int_{\Theta_{\text{crit}}}^{\Theta_{\text{tran}}}(P_i(\Theta)+P_{\text{spool}})d\Theta}_{\text{relief closed}}\right] \tag{9}$$

$$=\frac{Q_{\text{in}}}{\omega}\left[P_{\text{relief}}\Theta_{\text{crit}}+P_{\text{open}}\Theta_{\text{tran}}\left(\frac{\Theta_{\text{tran}}}{\Theta_{\text{crit}}}-1\right)+P_{\text{spool}}(\Theta_{\text{tran}}-\Theta_{\text{crit}})\right] \tag{10}$$

$$=\frac{2Q_{\text{in}}R_w}{\omega D}\sqrt{P_{\text{open}}}\left[2\sqrt{P_{\text{relief}}}-\sqrt{P_{\text{open}}}+P_{\text{spool}}\left(\frac{1}{\sqrt{P_{\text{open}}}}-\frac{1}{\sqrt{P_{\text{relief}}}}\right)\right] \tag{11}$$

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The energy lost when opening the inlet to load (interval b_2) is calculated similarly. The main difference is that when the relief valve is open, the flow through the relief valve is throttled across P_{relief}, whereas the flow through the rotary valve is throttled across $P_{\text{relief}} - P_{\text{load}}$. In angle coordinates, the instantaneous flow through the rotary valve (Q_i) with varying orifice area when the relief valve is open is

$$Q_i(\Theta) = Q_{\text{in}}\sqrt{\frac{P_{\text{relief}} - P_{\text{load}}}{P_{\text{open}}}}\frac{\Theta}{\Theta_{\text{tran}}} \tag{12}$$

The critical angle for the load transitions, Θ'_{crit}, occurs when $Q_i(\Theta'_{\text{crit}}) = Q_{\text{in}}$. Thus,

$$\frac{\Theta'_{\text{crit}}}{\Theta_{\text{tran}}} = \sqrt{\frac{P_{\text{open}}}{P_{\text{relief}} - P_{\text{load}}}} \tag{13}$$

The resulting energy loss during each load transition is

$$E_{\text{load}} = \frac{1}{\omega}\left[\int_0^{\Theta'_{\text{crit}}}\{(Q_{\text{in}} - Q_i(\Theta))P_{\text{relief}} + Q_i(\Theta) \times (P_{\text{relief}} - P_{\text{load}})\}d\Theta + \int_{\Theta'_{\text{crit}}}^{\Theta_{\text{tran}}} Q_{\text{in}}(P_i(\Theta) + P_{\text{spool}})d\Theta\right] \tag{14}$$

$$= \frac{2Q_{\text{in}}R_w}{\omega D}\sqrt{P_{\text{open}}}\left[\frac{2P_{\text{relief}} - 1.5P_{\text{load}} - P_{\text{spool}}}{\sqrt{P_{\text{relief}} - P_{\text{load}}}} + \frac{P_{\text{spool}} - P_{\text{open}}}{\sqrt{P_{\text{open}}}}\right] \tag{15}$$

The average transition power loss is derived from the total energy loss per cycle (four transitions) and the PWM frequency ($f_{\text{PWM}} = N * \omega/2 * \pi$)

$$\Pi_{\text{trans}} = (2E_{\text{load}} + 2E_{\text{tank}})\frac{f_{\text{PWM}}}{2\pi} \tag{16}$$

Using the definition in Eq. (16), the loss for the relief circuit is

$$\Pi_{\text{trans,relief}} = \underbrace{\kappa\sqrt{P_{\text{open}}}}_{\text{optimizable}}\frac{Q_{\text{in}}}{2}\left[\frac{2P_{\text{relief}} - 1.5P_{\text{load}} - P_{\text{spool}}}{\sqrt{P_{\text{relief}} - P_{\text{load}}}} + \frac{2(P_{\text{spool}} - P_{\text{open}})}{\sqrt{P_{\text{open}}}} + \frac{2P_{\text{relief}} - P_{\text{spool}}}{\sqrt{P_{\text{relief}}}}\right] \tag{17}$$

The loss for the check circuit is calculated similarly. The main difference is that the critical angle when the check valve first begins to open or close corresponds to a pressure of ($P_{\text{check}} + P_{\text{load}}$) for the tank transitions and P_{check} for the load transitions. The resulting loss is

$$\Pi_{\text{trans,check}} = \underbrace{\kappa\sqrt{P_{\text{open}}}}_{\text{optimizable}}\frac{Q_{\text{in}}}{2}\left[\frac{2P_{\text{check}} + 1.5P_{\text{load}} - P_{\text{spool}}}{\sqrt{P_{\text{check}} + P_{\text{load}}}} + \frac{2(P_{\text{spool}} - P_{\text{open}})}{\sqrt{P_{\text{open}}}} + \frac{2P_{\text{check}} - P_{\text{spool}}}{\sqrt{P_{\text{check}}}}\right] \tag{18}$$

Remarks.

(1) Equations (17) and (18) reveal that the average transition loss of the rotary valve is independent of PWM frequency. This is unlike linear valves with fixed transition time where transition losses increase with frequency.

(2) Separating these equations into two parts reveals that the transition loss is dependent only on geometric and system parameters: $\kappa\sqrt{P_{\text{open}}}$ consists of rotary valve design parameters that can be optimized to reduce losses, while the remaining terms consist mostly of system operating conditions and check and relief valve settings.

(3) The last term in Eqs. (17) and (18) highlights the loss saving advantage of the check circuit. In Eq. (17), P_{relief} is constrained by the load pressure and is typically $\gg P_{\text{spool}}$. In contrast, the last term of Eq. (18) is independent of load pressure, and P_{check} can be sized just slightly larger than $P_{\text{open}} + P_{\text{spool}}$. A comparison of how the transition losses compare between the two circuits is presented in Fig. 7 when $P_{\text{load}} < P_{\text{relief}}$.

4.2 Compressibility Losses. Compressibility loss consists of the energy per cycle that is required to compress the fluid in the inlet volume, V_{in} (see Fig. 2), from tank to load pressure. This energy is lost when the valve reopens to tank and is characterized by [2]

$$E_{\text{comp}} = V_{\text{in}}\int_{P_{\text{tank}}}^{P_{\text{load}}}\frac{P_{\text{in}}}{\beta(P_{\text{in}})}dP_{\text{in}} \tag{19}$$

P_{in} is the inlet volume pressure and $\beta(P_{\text{in}})$ is the pressure dependent bulk modulus of oil (used because of the substantial variation of bulk modulus with pressure between P_{tank} and P_{load}). The model used here, plotted in the top of Fig. 8, is the one proposed by Yu et al. [22] with the effect of air dissolving into oil neglected. This produces a more compressible model which yields conservative results. The power lost due to compressibility is found using the energy per unit volume needed to compress the oil (shown in the bottom of Fig. 8)

$$\Pi_{\text{comp}} = E_{\text{comp}}f_{\text{PWM}} \tag{20}$$

4.3 Leakage. Leakage paths exist in two locations (refer to Fig. 5): (1) across the helical land separating load pressure from tank and (2) across the spool ends (L_1) separating working fluid from the hydrostatic axial control chambers.

Because the outlet sections of the spool are always connected to load and tank pressure, the pressure differential across the helical land is nominally constant and equal to $P_{\text{load}} - P_{\text{tank}}$. Assuming laminar leakage flow [23], the leakage across the helical land is

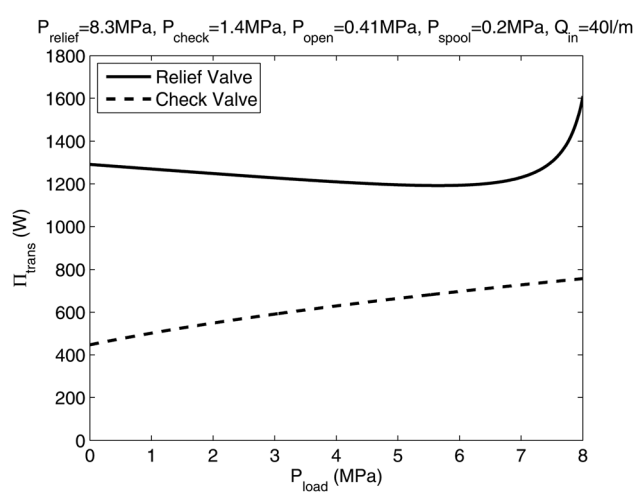

Fig. 7 Comparison of transition losses for relief and check circuits

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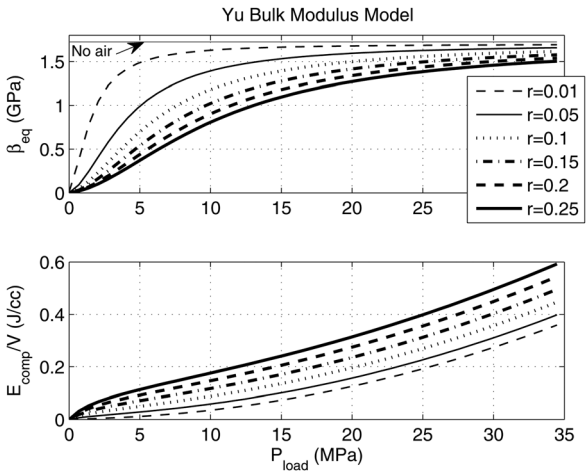

Fig. 8 Top: Yu pressure dependent bulk modulus for various fractions (*r*) of air entrainment and no dissolved air. Bottom: Unit compression energy required to compress fluid from P_{tank} to P_{load}.

$$Q_{\text{leak}} = \frac{L_p c_r^3 (P_{\text{load}} - P_{\text{tank}})}{12\mu R_w \sin\phi} \tag{21}$$

μ is the dynamic viscosity of hydraulic oil, c_r is the radial clearance between the spool and sleeve, $R_w \sin\phi$ is the thickness of the land normal to its edge, ϕ is the angle of the rhombus, and L_p is the perimeter of the leakage path parallel to the land edges

$$L_p = N\left(\frac{2L - R_h}{\sin\phi}\right) \tag{22}$$

$L = \frac{\pi D R_h}{2NR_w}$ is the total axial travel. The resultant power loss is

$$\Pi_{\text{leak}} = \frac{L_p c_r^3 (P_{\text{load}} - P_{\text{tank}})^2}{12\mu R_w \sin\phi} \tag{23}$$

Similarly, the power loss due to leakage across the spool end land assuming laminar leakage flow is

$$\Pi_{\text{leak},L1} = \frac{\pi D c_r^3 (P_{\text{load}} - P_{\text{axial}})^2}{12\mu L_1} \tag{24}$$

P_{axial} is the pressure in the axial control chamber acting on the spool end adjacent to the load side. This pressure is used for axial positioning and is dependent on the method used.

4.4 Self-Spinning Velocity Analysis. The dynamics of the spool rotation can be determined by summing torques on the spool

$$J\ddot{\theta} = \tau_{\text{in}} + \tau_{\text{out}} - \tau_f \tag{25}$$

where J is the mass moment of inertia of the spool, $\ddot{\theta}$ is the angular acceleration, τ_{in} and τ_{out} are the torques contributed by the inlet and outlet turbines, respectively, and τ_f is the resisting torque due to viscous friction. τ_f is estimated using Petroff's law [24] which assumes Newtonian shear stress between concentric cylinders with relative rotary motion

$$\tau_f = \frac{1}{4} A_{\text{eff}} \frac{\mu}{c_r} D^2 \omega \tag{26}$$

A_{eff} is the effective bearing surface area of the spool which is discussed in more detail in Sec. 4.4.1.

A diagram of the inlet and outlet turbines and their corresponding control volumes (CV) is shown in Fig. 9. The inlet turbine is modeled as a stationary CV that surrounds the spool and tangential rhombic sleeve ports (offset R_{in} from the axis of rotation). The tangential sleeve ports serve as the inlet to the CV and generate angular momentum in the fluid. The fluid exits the inlet turbine axially (with no angular momentum) through an internal channel with cross-sectional area A_{axial} that leads to the outlet turbine. Thus, all angular momentum generated by the inlets is transferred to the spool. Assuming steady incompressible flow and one dimensional inlets/outlets, the inlet turbine torque is

$$\tau_{\text{in}} = \sum_1^N (R_{\text{in}} \times v)_{IN} \dot{m}_{\text{in}} = \frac{\rho R_{\text{in}}}{A_{\text{in}} N} Q_{\text{in}}^2 \tag{27}$$

ρ is the density of hydraulic oil, v is the mean velocity of the fluid as it exits the inlet and enters the spool, $\dot{m}$ is the corresponding mass flow rate, and $(\cdots)_{IN}$ refers to the conditions at the inlet of the CV.

The outlet turbine is modeled as a CV that rotates with the spool (Fig. 9). Sleeve effects on the fluid are assumed small since the sleeve/turbine interface allows the fluid to exit the turbine unguided before accumulating the fluid at the outlet port downstream. Fluid enters the CV axially from the inlet turbine with no angular momentum. As the fluid is turned by the curved turbine blades (n blades with $n = N$ in the current design for simplicity), a reaction torque is generated on the spool. Assuming A_{out}, the flow area through the blades (see Figs. 5 and 9), is constant and offset by R_{out} from the axis of rotation, the outlet turbine torque is

$$\begin{aligned}\tau_{\text{out}} &= \sum_1^N (R_{\text{out}} \times (v - v_{CV}))_{\text{OUT}} \dot{m}_{\text{out}} \\ &= R_{\text{out}} \rho Q_{\text{in}} \left(\frac{Q_{\text{in}}}{A_{\text{out}} N} - R_{\text{out}}\omega\right)\end{aligned} \tag{28}$$

$v_{CV} = R_{\text{out}}\omega$ is the velocity of the CV at the tip of the blades where the fluid exits. Equating the inlet and outlet torques to the friction torque at steady state (Eq. (25)) produces the steady state angular spool velocity

$$\omega = \frac{4\rho}{ND^2\left(A_{\text{eff}}\frac{\mu}{c_r} + \frac{4R_{\text{out}}^2}{D^2}\rho Q_{\text{in}}\right)} \frac{R_{\text{in}}}{\bar{A}} Q_{\text{in}}^2 \tag{29}$$

where $\bar{A}$ is an equivalent combined flow area

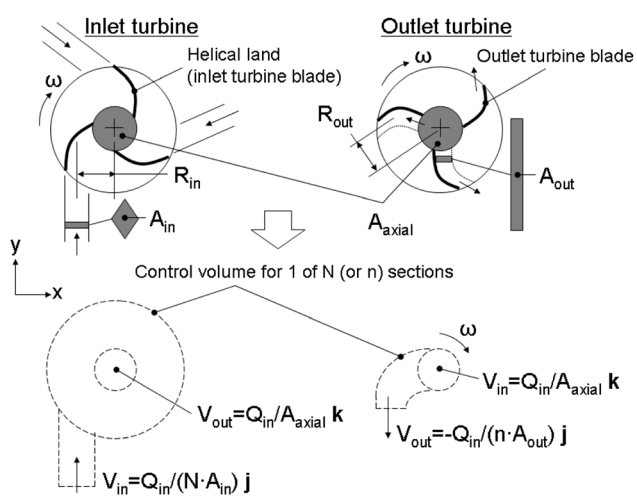

Fig. 9 Inlet and outlet turbines with their control volumes

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$$\frac{1}{\bar{A}} = \frac{1}{A_{\text{in}}} + \frac{R_{\text{out}}}{R_{\text{in}} A_{\text{out}}} \tag{30}$$

Remarks.

(1) Equation (28) illustrates competing effects in the outlet turbine. $Q_{\text{in}}/(A_{\text{out}}N)$ corresponds to the angular momentum generated by the fluid flow as it is turned by the turbine blades. However, this momentum is counteracted by the $R_{\text{out}}\omega$ term which reflects the angular momentum that must be transferred to the fluid as it is forced to rotate with the same circumferential velocity as the blades. Consequently, when the tip velocity of the outlet turbine exceeds the mean velocity of the fluid (i.e., $R_{\text{out}}\omega > Q_{\text{in}}/(A_{\text{out}}N)$), the turbine will act as a pump and require torque to maintain its velocity.

(2) For applications where low throttling is desired and a slower PWM frequency is acceptable, the inlet turbine alone can be used for self-spinning and the outlet section can be designed to minimize pressure drop. Equation (29) can be simplified to this case by setting $R_{\text{out}} = 0$ which reduces Eq. (25) to $\tau_{\text{in}} = \tau_f$ (both designs will be verified experimentally in Sec. 7.2)

$$\omega = \frac{4\rho}{ND^2 A_{\text{eff}} \frac{\mu}{c_r}} \frac{R_{\text{in}}}{A_{\text{in}}} Q_{\text{in}}^2 \tag{31}$$

4.4.1 Effective Bearing Surface and Friction Analysis. A method is developed in this section for estimating the total friction effects on the valve spool. This is done by finding an equivalent spool bearing surface area A_{eff} that can be used in Eq. (26). The first type of friction present is the journal bearing friction which is due to the helical land and L_1/L_2 sealing lands which have a combined surface area of

$$A_b = \pi D[R_h + 2(L_1 + L_2)] \tag{32}$$

The second type of friction is due to fluid recirculation in the pockets between the helical lands and the outlet turbine blades with surface areas πDL and $2\pi DL_{\text{out}}$, respectively (refer to Fig. 5). By using CFD to characterize the friction in the pockets, the effect of this friction can be combined with A_b to form an effective bearing surface area

$$\begin{aligned} A_{\text{eff}} &= A_b + \lambda[\alpha_{\text{in}}\pi DL + \alpha_{\text{out}} 2\pi DL_{\text{out}}] \\ &= \pi D[R_h + 2(L_1 + L_2) + \lambda(\alpha_{\text{in}} L + 2\alpha_{\text{out}} L_{\text{out}})] \end{aligned} \tag{33}$$

α_{in} and α_{out} are the ratios of shear stress due to fluid recirculation (σ_p) to the bearing surface shear (σ_b), and λ is a correction factor used to match the predicted self-spinning velocity to experimental data. σ_p is estimated from a CFD code that assumes steady two-dimensional incompressible Newtonian flow. The spool geometry is approximated by a rectangular chamber with moving upper boundary (Fig. 10). The CFD analysis shows that a single trend line (Fig. 11) is able to capture the dependence of normalized shear $\left(\sigma_p/(\rho V^2)\right)$ on aspect ratio ($\xi = w/h$) and Reynolds number ($\text{Re} = \rho V h/\mu$)

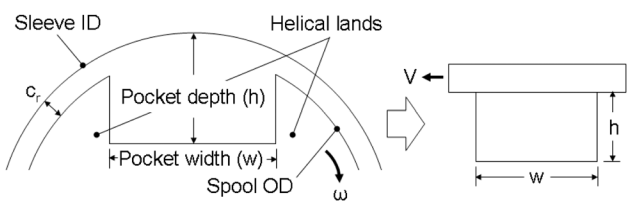

Fig. 10 Pocketed volume and corresponding CFD model

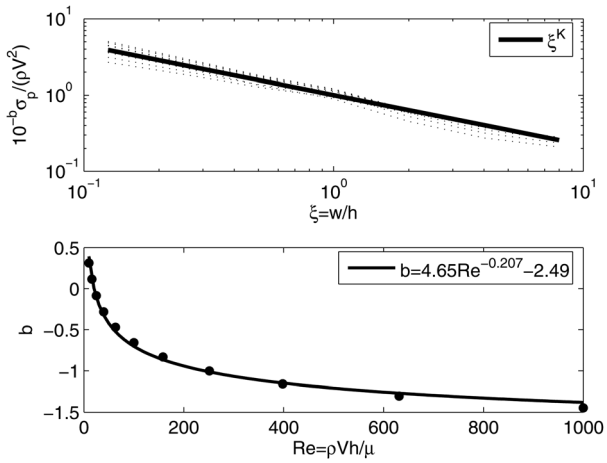

Fig. 11 Normalized pocket shear ($K = -0.653$)

$$\frac{\sigma_p}{\rho V^2} = 10^b \xi^K \tag{34}$$

b is a function of Re and is defined at the bottom of Fig. 11 and $K = -0.653$. w is the chamber width, h is the chamber depth, and $V = R\omega$ is the sliding velocity of the upper boundary corresponding to the spool's rotational velocity ω.

Equation (34) is used to quantify the shear stress due to recirculation in the fluid pockets of the nonbearing surface area. Because the aspect ratio of the pocket between the helical lands of the inlet turbine varies with respect to the spool's axial position, an average shear stress is defined for the inlet turbine. Integrating the differential shear stress along the length of the pocket and then dividing by the total surface area ($0.5\bar{w}L$, where $\bar{w} = \pi D/N$ is the widest portion of the pocket) and knowing that the aspect ratio varies linearly according to $\xi(l) = \bar{w}/(Lh)l$ for $0 \le l \le L$ produces

$$\begin{aligned} \sigma_{p,\text{avg}} &= \frac{\int_{l=0}^{l=L} \sigma_p w dl}{0.5\bar{w}L} = \frac{\int_{l=0}^{l=L} 10^b \rho V^2 \xi(l)^K \xi(l) h dl}{0.5\bar{w}L} \\ &= \frac{2}{K+2} \rho V^2 10^b \bar{\xi}^K \end{aligned} \tag{35}$$

$\bar{\xi} = \bar{w}/h$ is the largest aspect ratio corresponding to the widest portion of the pocket. Using Eq. (35), $\alpha_{\text{in}} = \sigma_{p,\text{avg}}/\sigma_b$. Conversely, because the aspect ratio of the fluid pocket between the outlet turbine blades is constant, the shear stress is calculated directly from Eq. (34) and $\alpha_{\text{out}} = \sigma_p/\sigma_b$.

The journal bearing shear σ_b is estimated using the shear stress equation for Newtonian flow between two moving plates with constant relative velocity [25]

$$\sigma_b = \mu \frac{R\omega}{c_r} \tag{36}$$

Figure 12 presents the predicted driving power needed to overcome viscous friction. Only 30 W is required to achieve 100 Hz PWM frequency with the prototype valve described in Table 1 and Sec. 6. $\alpha_{\text{in}} \approx 0.05$ and $\alpha_{\text{out}} \approx 0.035$ for the prototype, and the corresponding drag torque due to recirculation is about 30% of the total friction torque.

4.5 Output Flow Analysis. Due to the rotary valve's nonzero switching time, flow is bypassed through the relief or check valve during transition to limit pressure spikes. Consequently, the valve's normalized axial position does not correspond to the true duty ratio of the valve (i.e., 50% travel does not equal 50% VVDP

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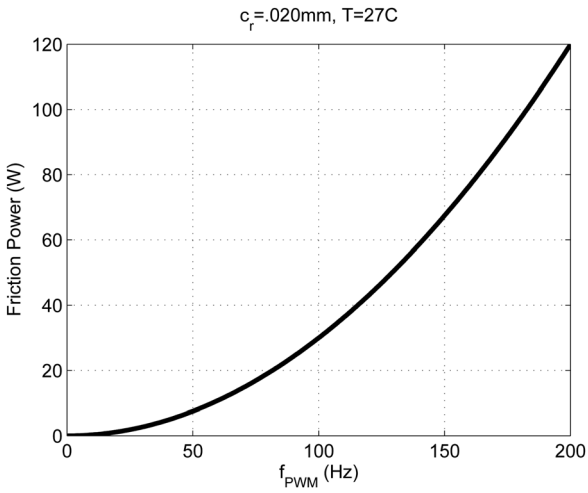

Fig. 12 Power required to overcome viscous friction

Table 1 Rotary valve parameters

	Prototype	Opt. 15 Hz	Opt. 75 Hz
D (mm)	25.4	31	17.8
L_s (mm)	98.0	127[a]	127[a]
R_w (mm)	3.7	8.1	4.7
R_h (mm)	6.5	13.3[a]	13.3[a]
L_1 (mm)	3.2	2	1.6
NA_{in} (mm^2)	36.5	161.6	93.8
NA_{out} (mm^2)	140.5	120.0	93.6
P_{open} (MPa)	0.41	0.021	0.062
P_{spool} (MPa)	0.205	0.081	0.318
c_r (mm)	0.025	0.020	0.023
κ	0.56	1.0[a]	1.0[a]
V_{sleeve} (cc)	61	9.2	7.9
V_{in} (cc)	72.4	20.5	19.3

[a]The parameter is at the bound specified in the optimization.

output flow). The two can be correlated by finding the volume of fluid bypassed through either the relief (Fig. 2(*a*)) or check valve (Fig. 2(*b*)) during transition. Using the approach and definitions found in Sec. 4.1.2 and integrating with respect to spool angle, the bypassed fluid volume for a single transition is

$$V_{by} = \frac{1}{\omega}\int_{\Theta=0}^{\Theta=\Theta_{\text{crit}}} (Q_{\text{in}} - Q_i(\Theta))d\Theta = \frac{Q_{\text{in}}R_w}{\omega D}\sqrt{\frac{P_{\text{open}}}{\Delta P}} \tag{37}$$

$Q_i(\Theta)$ is the flow through the rotary valve when the relief or check valve is open and ΔP is the critical angle pressure differential when the relief or check valve closes.

For the relief circuit, the volume of fluid bypassed during the two load transitions decreases the output flow of the system. On the other hand, the two tank transitions have no effect. Accounting for two load transitions and substituting $\Delta P = P_{\text{relief}} - P_{\text{load}}$, κ from Eq. (2), and multiplying by the PWM frequency, the average flow bypassed is

$$Q_{\text{relief}} = \kappa\frac{Q_{\text{in}}}{4}\sqrt{\frac{P_{\text{open}}}{P_{\text{relief}} - P_{\text{load}}}} \tag{38}$$

For the check circuit, the bypassed flow is ported to load regardless of whether the rotary valve is transitioning to load or tank. Therefore, the bypassed flow during the two tank transitions will increase the output flow. $\Delta P = P_{\text{check}} + P_{\text{load}}$ for the tank transitions and the average bypassed flow is

$$Q_{\text{check}} = \kappa\frac{Q_{\text{in}}}{4}\sqrt{\frac{P_{\text{open}}}{P_{\text{check}} + P_{\text{load}}}} \tag{39}$$

As with the transition loss, the bypassed flow through the relief or check valve is independent of PWM frequency and dependent only on geometric and system parameters. Using Eqs. (38) and (39), the duty ratio (VVDP output flow fraction) with respect to the valve's normalized axial position z is

$$s_{\text{relief}} = z - \frac{Q_{\text{relief}}}{Q_{\text{in}}} \tag{40}$$

$$s_{\text{check}} = z + \frac{Q_{\text{check}}}{Q_{\text{in}}} \tag{41}$$

For the prototype described in Table 1 and Sec. 6, $Q_{\text{relief}}/Q_{\text{in}} = 0.077$ and $Q_{\text{check}}/Q_{\text{in}} = 0.031$ at a load pressure of 6.9 MPa.

4.6 Trade Off and Design Summary. Several design trade offs exist that can be exploited to improve the rotary valve's efficiency. Using the geometric constraints that the sides of the rhombic inlets are parallel to the helical lands and the valve is critically lapped produces the equality $R_h/R_w = L/\pi D/2N$. Using this equality and substituting in κ and P_{open} from Eqs. (2) and (3) produces the constraint

$$\kappa^2\sqrt{P_{\text{open}}} = \frac{\sqrt{128\rho}Q_{\text{in}}}{\pi C_d DL} = \text{constant} \tag{42}$$

Equation (42) states that P_{open} and κ cannot decrease simultaneously for a fixed DL. However, since transition losses (Eqs. (17) and (18)) scale with $\kappa\sqrt{P_{\text{open}}}$, Eq. (42) suggests that κ should be increased in order to decrease $\kappa\sqrt{P_{\text{open}}}$.

Another trade off exists between spool velocity, spool size, leakage, and flow area (pressure drop). Making the simplifying assumptions that $R_{\text{in}} \approx R_{\text{out}} \approx D/2$ in Eq. (29) and $A_{\text{in}} \approx A_{\text{out}}$ in Eq. (30) yields

$$\omega \propto \frac{c_r Q_{\text{in}}^2}{D}\left(\frac{1}{A_{\text{in}}}\right) \propto \frac{c_r Q_{\text{in}}\sqrt{P_{\text{open}}}}{D} \tag{43}$$

Equation (43) shows that ω can be increased at the cost of throttling by increasing P_{open}, at the cost of leakage by increasing c_r, or by reducing the spool's diameter D which reduces surface area and friction moment arm. Equations (42) and (43) suggest that the self-spinning velocity can be increased without a penalty in transition loss by increasing the spool length L in Eq. (42) to compensate for a decrease in diameter. An additional motive for decreasing D at high speeds, which was not derived explicitly in Sec. 4.2, is that the volume of the inlet pressure rail (shown in Fig. 3) scales with D, which accounts for a majority of the compressible inlet volume. Therefore, a smaller diameter spool has the additional benefit of decreasing compressibility losses, which are important at high PWM frequencies and load pressures.

Table 1 contains the geometric parameters describing the prototype valve and designs optimized for 15 Hz and 75 Hz PWM frequency at $P_{\text{load}} = 6.9$ MPa and $Q_{\text{in}} = 40$ lpm using the design equations derived in Secs. 4.1–4.4 (see Ref. [26] for details on the optimization method). Note that the total compressible volume, V_{in}, is the volume in the sleeve, V_{sleeve}, plus one half of the pump displacement. As expected, the optimization algorithm drives $\kappa \to 1$ and L_s, the total spool length, to its upper bound while decreasing D as the desired PWM frequency is increased from 15 to 75 Hz.

Table 2 contains a breakdown of the various forms of loss calculated using the design equations for the prototype and optimized valves. In both cases, the transition loss is the dominant form of loss, accounting for as much as 77% of the total losses for the relief circuit and about 67% of the losses for the check circuit. Using the parameters from the optimization for 15 Hz reduces all

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Table 2 Breakdown of VVDP losses using the parameters $P_{load} = 6.9$ MPa, $Q_{in} = 40$ lpm, $r = 0.10$. Total output power is 4.6 kW.

	Prototype 15 Hz	Opt. 15 Hz
$\Pi_{trans,relief}$ (W)	1222	567
$\Pi_{trans,check}$ (W)	725	368
Π_{open} (W)	179	—
Π_{comp} (W)	69	20
$\Pi_{leak} + \Pi_{leak,L1}$ (W)	113	55

of the losses. Fully open losses no longer exist in the optimized case when $\kappa = 1$ because the spool is always in transition.

5 Dynamic System Simulations

A dynamic model of a self-spinning rotary valve based VVDP, including the effects of the compressible inlet volume, accumulator, and transition/full open throttling, is developed in this section. Several approximations are used to simplify the model: the accumulator gas is assumed to behave adiabatically, line losses are neglected, and the relief or check valve in the circuit opens instantaneously. The pressure drop across the spool, P_{spool}, is also assumed constant.

The VVDP is modeled with two states. The states include the inlet volume pressure P_{in}, governed by compressible oil dynamics, and the load pressure P_{load}, governed by accumulator gas dynamics. Using the definition of bulk modulus, the dynamics of the oil volume can be derived as

$$\dot{P}_{in} = \frac{\beta(P_{in})}{V_{in}} Q_{vol} \tag{44}$$

V_{in} is the inlet volume and Q_{vol} (defined in Fig. 2) is the net flow into the inlet volume found by summing all of the input and output flows to the volume. $\beta(P_{in})$ is the pressure dependent bulk modulus described in Sec. 4.2.

The dynamics of the accumulator gas are derived from the ideal gas adiabatic compression equation

$$\dot{P}_{load} = \gamma \frac{P_{load}^{(1+1/\gamma)}}{P_0^{1/\gamma} V_0} Q_{acc} \tag{45}$$

P_0 is the gas precharge pressure, V_0 is the initial gas volume, Q_{acc} is the net flow of oil into the accumulator (Fig. 2), and $\gamma = 1.4$ (the ratio of specific heats for air and nitrogen). Equations (44) and (45) are simulated using the MATLAB/SIMULINK software package.

6 Experimental Hardware

The prototype rotary valve components (pump housing, valve sleeve, and spools) are shown in Fig. 13. The VVDP test stand consists of a 5.6 kW ac motor driving a 22.8 cc fixed-displacement vane pump modified for use with the rotary valve. The ac motor is controlled with a variable-frequency drive for testing at different flow rates. Pressure sensors and a cartridge relief or check valve are mounted on the sleeve. The sleeve contains two output ports. One port returns to tank. The other port connects to a 16 l diaphragm accumulator precharged to 2.1 MPa, a flow meter, an oil filter, and a needle valve load. During self-spinning operation, a small gerotor pump is used to control the spool's axial position. A special spool with shaft extension (see Fig. 13) is used to control the spool speed externally. This spool is used to characterize the valve at different PWM frequencies for a fixed flow rate. Mobile DTE 25 hydraulic oil is used in the test stand with $\rho = 876$ kg/m^3 and $\mu = 0.0387$ Pa s at 40 °C. During operation, both the amplitude and characteristic of the sound produced by the VVDP resemble that of a motorcycle or lawn mower, depending on the PWM frequency.

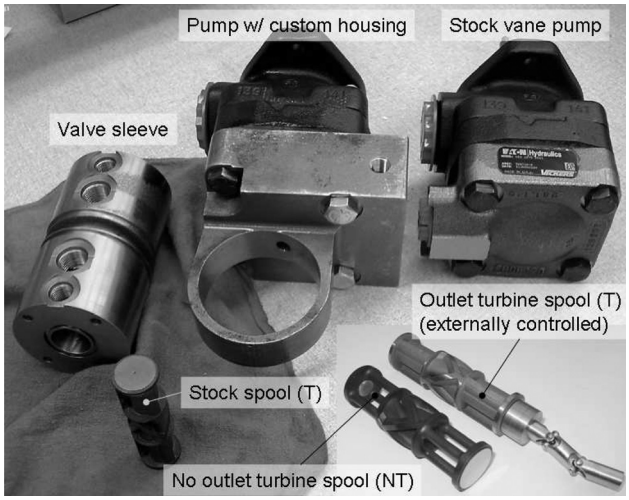

Fig. 13 Prototype rotary valve hardware

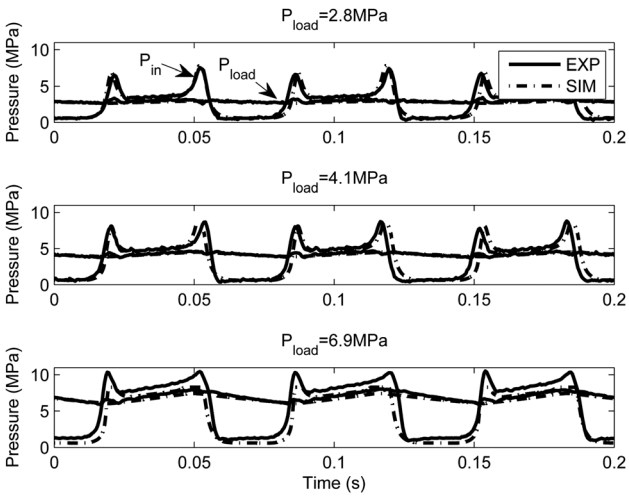

Fig. 14 15 Hz pressure profiles: 50% travel

7 Simulation and Experimental Results

Experimental data were acquired while operating the rotary valve between 15 and 75 Hz PWM frequency for a fixed axial position, load pressure, and input flow rate ($Q_{in} = 40$ lpm). $P_{relief} = 8.3$ MPa and $P_{check} = 1.4$ MPa.[2] Tests were run at a nominal oil temperature of 30 °C with data sampled at 2 kHz.

7.1 Pressure Profiles. Simulated and experimental pressure profiles are presented in Figs. 14–16. The simulated pressure profiles are matched to experimental data by tuning r, the fraction of entrained air. $r = 0.10$, or 10% air entrainment, was found to provide a reasonable match at 15 Hz PWM frequency. While this level of air entrainment is higher than the typical 2–7% seen in the literature [27] and via word of mouth in industry, it appears possible due to the test stand's unintentional poor reservoir design.[3]

Figure 14 shows good correlation between the simulation and experimental data at 15 Hz PWM frequency for various load pressures at 50% travel. Spool underlap was included in the simulation after the discovery of an unintentional 26% underlap in the

[2] P_{relief} and P_{check} are sized such that $P_{relief} = P_{load} + P_{check}$ at the maximum load pressure tested (6.9 MPa). $P_{check} > P_{open} + P_{spool}$.

[3] The test stand reservoir is unsealed, contains no baffling, and the return lines are not submerged and are located near the pump inlet. As a result, splashing occurs in the oil at the PWM frequency due to the tank line.

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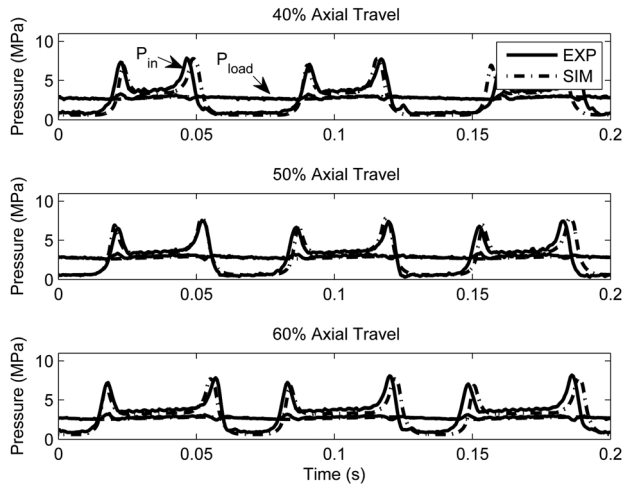

Fig. 15 15 Hz pressure profiles: $P_{load} = 2.8$ MPa

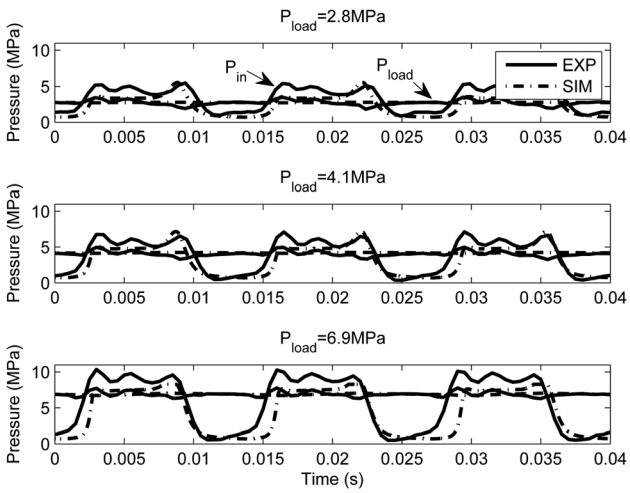

Fig. 16 75 Hz pressure profiles: 50% travel

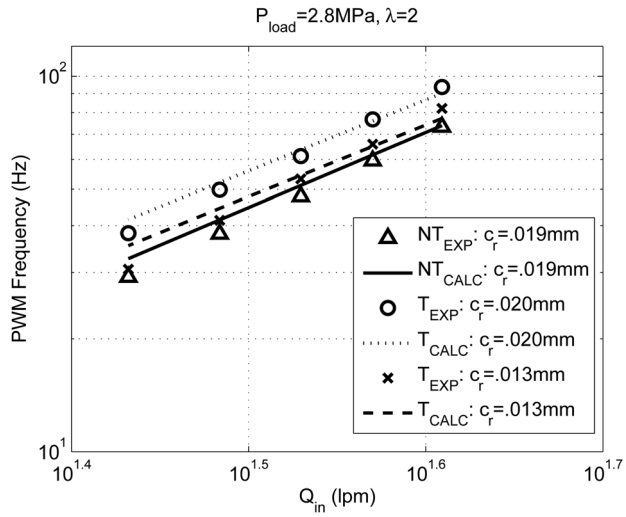

Fig. 17 PWM frequency versus Q_{in} (loglog)

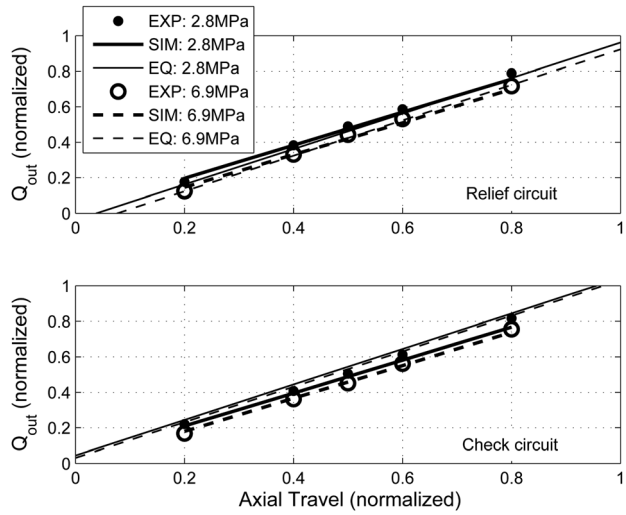

Fig. 18 Output flow versus axial position (15 Hz)

prototype. The underlap decreases the magnitude of the transition peak from load to tank providing a better overall match with the experimental data. The square-wave characteristic of P_{in} is clearly visible at 15 Hz indicating that the rotary valve is pulsing the flow. Figure 14 also shows that the full open pressure drop across the valve spool and sleeve is 0.62 MPa at 15 Hz and $Q_{in} = 40$ lpm (with $P_{open} = 0.42$ MPa and $P_{spool} = 0.2$ MPa). This is consistent with the prediction of 0.61 MPa from simulation and CFD. The variation of pulse width with axial position is shown in Fig. 15, thereby validating the helical land duty ratio concept.

At 75 Hz PWM frequency (Fig. 16), more deviation arises between the simulation and the experiment although a reasonable match is still achieved. More ripples are evident on the experimental inlet pressure when the on/off valve is connected to the load branch. This may be due to fluid inertia or water hammer effects, which are not modeled. The simulation is able to capture the increasing sluggishness of P_{in} due to compressibility at higher switching frequencies caused by the large inlet volume of the unoptimized prototype and the higher than usual air entrainment. This emphasizes the need for V_{in} to be small and the importance of proper reservoir design for efficient operation at high frequencies.

7.2 Self-Spinning Validation. The self-spinning concept is validated by controlling the spool's axial position hydrostatically. Using this method, the spool is completely isolated from the sleeve and spun solely with fluid forces. Two spool designs, one with outlet turbines (labeled T in Fig. 13 corresponding to Eq. (29)) and one without (NT, corresponding to Eq. (31)), were tested along with several clearances. Figure 17 confirms that ω is proportional to Q_{in}^2 and compares the frequencies achieved experimentally with the prediction from the angular momentum analysis. A_{eff} (see Eq. (33)) was calculated with α_{in} and α_{out} based on a spool frequency of 25 Hz using a correction factor of $\lambda = 2$ on these nonbearing surface area shear ratios. Friction due to fluid recirculation in the bladeless outlet section of the NT spool was found to be significant as including α_{out} produced a better match with the experiment (NT_{CALC} in Fig. 17).

Figure 17 reveals that the inlet turbine contributes a majority of the torque used to spin the spool. The NT spool spins on average 23% slower than a T spool of similar clearance. Decreasing clearance has a similar effect. Between the loose clearance ($c_r = 0.020$ mm) and tight clearance ($c_r = 0.013$ mm) T spools, the tight clearance spool spins on average 15% slower. Additional experimental data show that the PWM frequency is independent of axial position, with less than 15% variation in PWM frequency between the normalized output flows of 0.4–0.7.

7.3 Flow Modulation. Figure 18 compares the predicted, simulated, and experimental axial position/output flow relationship for both the relief (top) and check (bottom) circuits. In the relief

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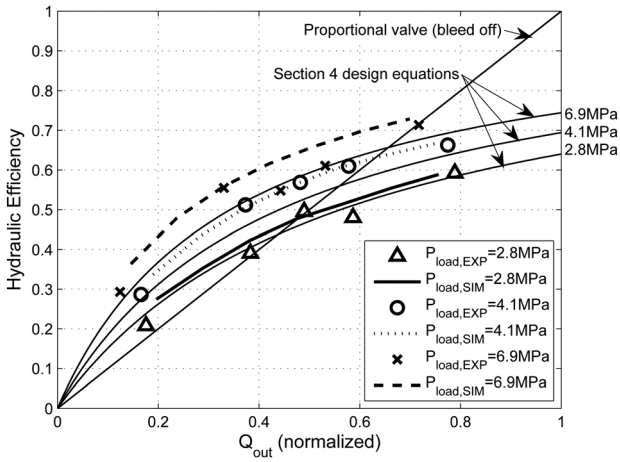

Fig. 19 Efficiency at 15 Hz: relief

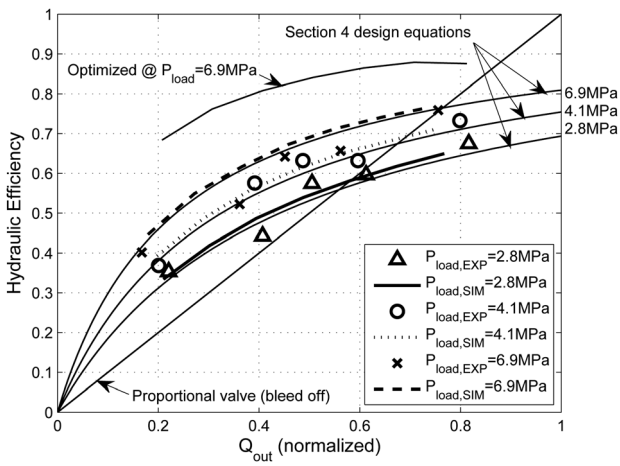

Fig. 20 Efficiency at 15 Hz: check

circuit, increasing P_{load} decreases Q_{out} for a given axial position. Both the analysis and simulation show that as P_{load} approaches P_{relief}, the relief valve is open for a greater part of each cycle since P_{in} reaches P_{relief} earlier in the transition. At $P_{\text{load}} = 2.8$ MPa, the simulation predicts that the relief valve is open for roughly 1.5 ms, while this time increases to 9 ms when $P_{\text{load}} = 6.9$ MPa.

In the check circuit, the duty ratio is greater than the corresponding normalized axial position z (i.e., $Q_{\text{out}} = sQ_{\text{in}} > zQ_{\text{in}}$) as predicted in Sec. 4.5, although only at low load pressure. The inconsistency at high load pressure is likely due to compressibility and valve underlap. Compressibility during the tank transitions causes the check valve to open later in the transition, thus leading to less flow bypassed to load. Valve underlap also decreases the flow to load by introducing a leakage path between load and tank.

7.4 Hydraulic Efficiency. The hydraulic efficiency of the rotary valve is calculated by comparing the input hydraulic power to the valve to the output hydraulic power of the system[4]

$$\eta = \frac{\left(\sum_{i=1}^{n} P_{\text{load}}(i) Q_{\text{out}}(i) \Delta t\right) / T_{\text{avg}}}{\left(\sum_{i=1}^{n} P_{\text{in}}(i) Q_{\text{in}} \Delta t\right) / T_{\text{avg}}} \tag{46}$$

$n = T_{\text{avg}}/\Delta t$ is the number of time steps over which the summation is performed. Experimental data are averaged over 10 s with $\Delta t = 0.5$ ms, while simulation data are averaged over 1 s with $\Delta t = 0.01$ ms. Since leakage is present in the measured flow, it is included in the simulation results. Pump efficiency, however, is not included.

Figures 19 and 20 compare the experimental and simulated VVDP efficiencies at 15 Hz PWM frequency with the characteristic efficiency of an equivalent bleed off system (Fig. 1). A good match is observed between the experiment and the dynamic models (refer to Sec. 5). In contrast, the efficiency predicted by the analytical design equations in Sec. 4 is noticeably lower for the relief circuit (Fig. 19). This is likely due to the dynamic coupling between compressibility and throttling that is not captured by the design equations, where the two are considered separately. In the actual system (and dynamic model), compressibility slows down the pressure dynamics, i.e., increases the rise and fall time of the inlet pressure during valve transition. Because incompressibility is assumed in the orifice equation used in the transition loss analysis (Sec. 4.1.2), the analysis predicts that the relief and check valves open sooner in the transition than they do when compressibility is included. Therefore, transition losses are overpredicted in the relief circuit since the maximum throttling occurs at the relief pressure. In the check circuit, however, the throttling losses across the check valve are small so there is less impact on efficiency. At 15 Hz, the relief circuit exhibits up to 25% efficiency improvement over the bleed off system for high load pressures and displacements less than 70% (Fig. 19). At high displacements, the rotary valve is less efficient due to P_{open} and P_{spool} (see Sec. 4.1.1), which must be reduced to achieve high efficiency at high displacement. Full displacement efficiency can be further improved by eliminating the transition losses at full displacement in the prototype valve. This can be done by increasing the length of the center section of the valve (L_{in} in Fig. 5) without modifying the helical lands.

By switching to the check valve configuration shown in Fig. 2(*b*), the efficiency can be further improved up to 5% across a full range of displacements and load pressures (Fig. 20). This increase in efficiency is accomplished by limiting transition throttling to $P_{\text{check}} + P_{\text{load}}$ and also by porting the pressurized bypass flow to load rather than tank. This configuration demonstrates higher efficiency than the proportional valve system for displacements under 75%.

The optimized curve in Fig. 20 projects the potential efficiency of the check circuit using the optimized geometry in Table 1. Other improvements include reducing the check valve cracking pressure and eliminating spool underlap. Using these enhancements, the simulation predicts up to 22% efficiency improvement including 84% efficiency at 50% displacement. Another 5–10% improvement in efficiency can potentially be achieved with "soft-switching" (see Refs. [18,19]) which reduces transition losses by providing an alternate flow path.

Upon increasing f_{PWM} from 15 to 75 Hz (Fig. 21), there is more inconsistency between the experiment and the simulation. This is due to model mismatch which is observed in the pressure profiles shown in Fig. 16. At 75 Hz, the VVDP is only marginally more efficient than the bleed off circuit at higher load pressures and low displacements due to fluid compressibility in the unoptimized prototype. Compressibility increases throttling as mentioned before by slowing the dynamic response of P_{in} such that it no longer transitions as sharply between load and tank pressure. The effect is even greater at high load pressure. If P_{load} is too high, P_{in} may be too sluggish to reach the intended low pressure fully off state. However, upon decreasing V_{in} from 72.4 to 19.3 cc using CFD and applying the check circuit (with appropriately sized cracking pressure) and optimized geometry, the efficiency of the VVDP can be improved by up to 27% across a full range of displacements at 75 Hz. The optimized simulation predicts an efficiency of 77% at 50% displacement.

7.5 Volumetric Efficiency. The volumetric efficiency defined in this section is based on the leakage across the helical

[4]Efficiency is given on an absolute scale (0–1) rather than percentages.

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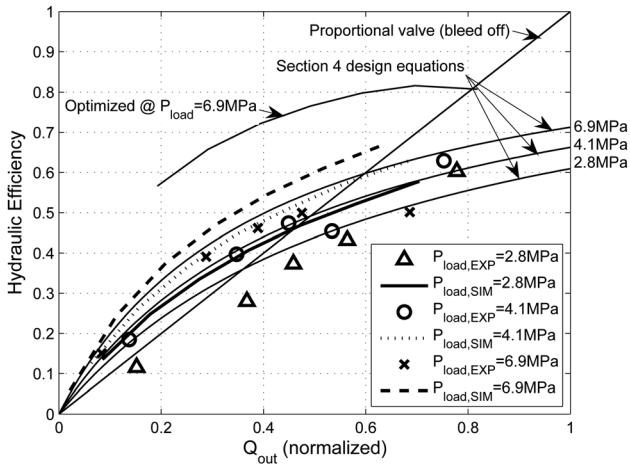

Fig. 21 Efficiency at 75 Hz: relief

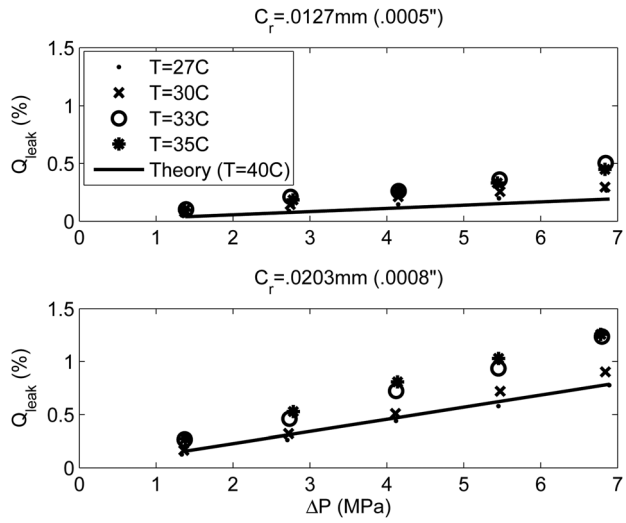

Fig. 22 Leakage across helical land

land of the rotary valve separating load pressure from tank (Sec. 4.3). Leakage across the end land (L_1) was not measured because it varies significantly depending on the axial positioning method used. Leakage across the helical land was measured experimentally by holding the spool position fixed with the inlet orifices fully open to load such that ΔP across the land was $P_{\text{load}} - P_{\text{tank}}$. Each test consisted of measuring the leakage for P_{load} between 1.4 and 6.9 MPa. Figure 22 provides a comparison between the measured leakage and the prediction from Eq. (21). The temperatures cited represent the average temperature over a single test. Leakage was measured to be within a factor of 3 of the prediction for oil temperatures up to 35 °C and the results indicate a strong dependence on oil temperature. For a system flow rate of $Q_{\text{in}} = 40$ lpm, Q_{leak} was roughly 0.5% for the tight clearance spool and less than 1.5% for the loose clearance. At $P_{\text{load}} = 6.9$ MPa, this corresponds to a volumetric efficiency greater than 98.5%.

8 Conclusions

A novel three-way self-spinning rotary on/off valve designed for pulse-width-modulation of hydraulic flow has been presented. A prototype valve based on the proposed design has simultaneously achieved high flow (40 lpm), low full open pressure drop (0.62 MPa), high speed (100 Hz self-spinning PWM frequency with 2.8 ms transition time), and low actuation power (30 W). With self-spinning, the entire actuation power is scavenged from fluid forces and no external power source is required.

A basic model of the valve suitable for design purposes and optimization has been developed. A two-state dynamic model has also been developed and simulated. Both models have been validated by collecting experimental data from a prototype valve utilized in a virtually variable displacement pump. The design equations and dynamic model show good agreement with experimental data at 15 Hz PWM frequency. However, as the frequency is raised to 75 Hz, the model shows less consistency with experimental data.

The mean output flow of the VVDP has been verified to exhibit a linear relationship with the rotary valve's axial position, thereby validating the helical land concept utilized in the spool. In addition, the self-spinning PWM frequency, predicted to be proportional to Q_{in}^2, has been confirmed by experiment. A correction factor of 2 on the nonbearing surface area (predicted simplistically) is needed to achieve a good match between the model and the experimental results.

The prototype VVDP has demonstrated efficiency improvements in comparison to an equivalent bleed off system over a large range of displacements at 15 Hz PWM frequency. At 75 Hz, compressibility negates any efficiency improvement due to the large 72.4 cc inlet volume of the prototype valve. A CFD analysis of the valve has shown that the inlet volume can be realistically reduced to 19.3 cc without a significant increase in throttling. Combining the reduced inlet volume with parameter optimization, the validated model predicts that an optimized VVDP at 50% displacement is able to achieve 84% efficiency at 15 Hz and 77% efficiency at 75 Hz. Proper reservoir design can realistically decrease the entrained air to more common values (2–7%) which would further improve efficiency.

Future work will focus on validating the optimization, demonstrating the VVDP with realistic loads, extending the rotary valve to the control of motors, and exploring the noise and vibration attributes of the valve.

Acknowledgment

This material is based upon the work supported by the National Science Foundation under Grant Nos. EEC-0540834 and ENG/CMS-0409832.

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